

Partha Ghosh

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I build LLM agents and production ML that sit on real systems—not demo chatbots. At **ConneqtedAgents** I work on **Optiify** (LangGraph tool-calling, MCP live point data, Neo4j graphs, and RAG) and on **Quantiify** (YOLOv8 + SAHI for drawing symbol count and legend detection). Previously at **REVE Systems** I shipped RAG chat using GraphRAG, LightRAG, multimodal RAG, agentic RAG, and LangChain, Bengali STT/TTS (**Kotha**, **Uccharon**), Temporal Fusion Transformer forecasts and a CatVTON virtual try-on for **Le Reve**.

EDUCATION

Shahjalal University of Science and Technology

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering

CGPA: 3.35 / 4.00 (Last 4 semesters: 3.61)

Feb 2018 – Mar 2023

WORK EXPERIENCE

ConneqtedAgents

Dec 2025 – Present • Remote

AI Engineer

- Built an in-process **LangGraph** agent for **Optiify** chat so building users can book after-hours access, inspect HVAC faults, and ask equipment questions in one conversation.
- Designed tool-calling workflows (booking, **FDD**, work orders, building context) with **FastAPI**, **DynamoDB**, and **InfluxDB** instead of stuffing raw building data into the prompt.
- Implemented an **MCP** agent for live point-condition time series: catalog lookup in **DynamoDB**, per-point **Influx** queries and prompt-led date windows.
- Built a **Neo4j** knowledge graph of HVAC equipment hierarchy, with **APOC**/full-text tools so the agent can search and traverse.
- Added document **RAG** over building functional docs and onboarding FAQs to ground fault explanations in site-specific knowledge.
- Trained **YOLOv8** with **SAHI** for **Quantiify** to detect legends and count symbols on large construction drawings without missing small objects.

REVE Systems

Jan 2024 – Dec 2025

Machine Learning Engineer

- Built a **RAG** chatbot for **REVE Chat** using **GraphRAG**, **LightRAG**, multimodal **RAG**, agentic **RAG**, and **LangChain**, with hybrid search (**BM25** + dense) for stronger responses.
- Fine-tuned **wav2vec 2.0** and **Whisper** for Bengali **STT** (**Kotha**) with **PEFT-LoRA**; evaluated on multiple test sets, including augmented data, to find model weaknesses.
- Fine-tuned a **VITS** model for Bengali **TTS** (**Uccharon**).
- Developed a voice-based banking agent with an **STT**-chatbot-**TTS** pipeline to automate customer conversations.
- Compressed models with quantization and **ONNX** for faster load/inference and offline use.
- Implemented face liveness detection for the NTMC biometric authentication project using **OpenCV**, **CNNs**, and **MediaPipe** (anti-spoofing).
- Developed a **Temporal Fusion Transformer (TFT)** pipeline for **Le Reve** to forecast production quantity and sales over multiple horizons, with attention-based interpretability so planners could see which product, season, and store signals drove each prediction.
- Built a virtual try-on for **Le Reve** with **CatVTON**, a lightweight diffusion model, so customers could see garments on a person image without a heavy try-on stack.
- Developed a chatbot with **BERT** and **Bi-LSTM**, and improved the **REVE Chat** dashboard frontend (**React.js**).

Codelab FZC

Jun 2023 – Jan 2024

Software Engineer

- Developed a 2D object-detection game and a typing game (players type sentences against a time limit) in **Unity** for Android and iOS.
- Contributed to an admin panel, implementing CRUD operations in **Laravel**.

TECHNICAL SKILLS & TECHNOLOGIES

Languages	Python, Java, C, C++, PHP, C#
Agentic AI	LangGraph, LangChain, LlamaIndex, MCP, Agentic RAG, OpenAI, Groq
Machine Learning	PyTorch, TensorFlow, OpenCV, Scikit-learn, MLflow, CI/CD
Web Technologies	Django, FastAPI, JavaScript (React.js, Node.js, Vue.js), HTML/CSS, Laravel
Databases	MySQL, MongoDB, Qdrant, Memgraph, Neo4j, InfluxDB, DynamoDB, Firebase
Cloud / Infra	AWS, Docker, Linux, Git, Bash
Tools	L ^A T _E X, Microsoft Office
Problem Solving	600+ problems — Codeforces (partha361), LeetCode (ppartha), LightOJ (parthaghoshu), StopStalk (ppaarthaa)

RESEARCH EXPERIENCE

- **Development of a Large Corpus of Spontaneous Speech for Improved Speech Recognition in Bangladeshi Bangla:** Created a 200-hour Bangladeshi Bangla spontaneous speech corpus and fine-tuned multilingual **wav2vec 2.0**, comparing it with a reading corpus (**SUBAK.KO**) to improve **ASR**. 2023–2024.
- **BanSpeech: A Multi-Domain Bangla Speech Recognition Benchmark Toward Robust Performance in Challenging Conditions:** Assessed a Bangla **ASR** model on **BanSpeech**, a 7.2-hour benchmark; it struggled with spontaneous speech and high OOV rates, highlighting domain selection. Published in *IEEE Access*, vol. 12, pp. 34527–34538, 2024.
Paper: ieeexplore.ieee.org/document/10453554.

PROJECTS

SUST Student Management System	
Web app for student and faculty records, departments, courses, and results. Admins add or update people and assign faculty to courses; faculty enter marks and students view their results.	
Technologies: Django, Python, HTML, CSS, Bootstrap, MySQL	
Automatic Attendance System Using Face Recognition	
Streamlit web app that recognizes faces from a webcam and marks class attendance. Enrolled each classmate with 5 photos so the model could identify people in front of the camera and log who attended.	
Technologies: Python, OpenCV, Streamlit	
Shoesly — E-commerce Website	
Shoe store with User, Supplier, Admin, and Bank routes. Shoppers filter products and pay from a bank account; admin takes a commission and pays the supplier; suppliers manage stock and fulfill orders.	
Technologies: Node.js, React.js, MongoDB, Express.js	

ACHIEVEMENTS

- 1st Runner-up, AI for Bangla 2.0 (2023).

EXTRA-CURRICULAR ACTIVITIES

CSE Society, SUST	
General Secretary, Assistant General Secretary, Executive Member	2019 – 2023
SUST ACM Student Chapter	
Joint Secretary	2021 – 2023

REFERENCE

Dr. Mohammad Shahidur Rahman
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Shahjalal University of Science and Technology, Sylhet
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